drylab exercises for the Wenzel Separation Science unit

Revision 3 — written against drylab v0.24.1. Adds Exercise 2A (van Deemter mechanism builder) and Exercise 7 (column overload and peak asymmetry) on the new Band shape section, and fixes a Craig homework pair that revision 2's own instructor-key pass caught as unreachable (D = 1.5/2.5 needs n ≈ 600 transfers; the slider stops at 200). A companion **instructor key** (drylab_instructor_key_v1.md) carries measured expected values for every step of every exercise.

**Revision 2 note — Every scenario, retention order, and number below was verified against the shipped model (2026-08-17); instructor-note values carry the operating point they were measured at. The first revision's Ex 5 co-elution pair (Decane/2-Octanone on SPB-50) failed verification — those solutes never co-elute in the oven range — and has been replaced with a measured pair.

Basis: drylab v0.23.0 (HPLC isocratic/gradient with sweeps, band tracks, Rs/α results columns and CSV exports; Craig apparatus; GC isothermal/programmed with solute picker, Compare and Golay tabs and CSV exports; CZE) mapped against Wenzel's POGIL materials (Learning Objectives, In-Class Problems, Out-of-Class Problems, Instructor's Manual, Text).

Design philosophy. Wenzel's problems are guided inquiry: students confront a model, predict, then justify, in facilitated groups of 3–4. drylab is the *model-generating instrument*: every exercise follows a **Predict → Simulate → Reconcile → Extend** loop, where the prediction is committed on paper *before* the slider moves. The simulator never replaces the reasoning step — it adjudicates it.

Classroom presets. Each section's sidebar has a **Classroom preset** dropdown that seeds the scenario's chemistry (analytes, phase, temperature, %B, solvent B); instrument geometry and pneumatics stay at defaults, so a preset states the problem, never the solution. Presets are also URL-selectable, so each group can be handed a link:

Exercise	Preset	URL parameter
Ex 3 — Resolution triage	HPLC: acetophenone + p-chlorophenol, 50 %B	`?hplc=ex3`
Ex 4 — General elution (LC)	HPLC: all 22 compounds, 30 %B	`?hplc=ex4`
Ex 4 — General elution (GC)	GC: alkane ladder C7–C15, SLB-5ms, 120 °C	`?gc=ex4`
Ex 5 — Alkane ladder	GC: C8/C10/C12, SLB-5ms, 120 °C	`?gc=ex5a`
Ex 5 — Polarity trio	GC: Decane, 1-Octanol, 2-Octanone, SLB-5ms, 120 °C	`?gc=ex5b`
Ex 5 — Co-elution hunt	GC: Tridecane + 2-Octanone, Wax, 120 °C	`?gc=ex5c`

Parameters combine (`?hplc=ex3&gc=ex5b`); unknown keys fall back to Default.

Interpretation flags (Brian):

1. "Key aspects of chromatography" interpreted as the topics Wenzel's unit *and* drylab both cover. Wenzel topics drylab cannot simulate — eddy diffusion (the GC model is open-tubular Golay with A = 0, itself a teachable line matching Wenzel's LO that open tubular columns

show no eddy diffusion; the HPLC a-term is fixed), steric exclusion, ion exchange, and the wet-lab extraction schemes — are noted in the coverage map rather than forced.
2. Six exercises spanning all four drylab sections, each with a conceptual in-class leg and a quantitative out-of-class leg. Trim at will.
3. Instructor-facing, with student-facing prompts inline in quotes. Expected outcomes carry measured values with operating points, so answer keys can be regenerated from the sim itself.

Exercise 1 — From separatory funnel to column: the Craig apparatus as the missing link

Wenzel linkage: Extraction problem set → theoretical plate / HETP objectives. The unit jumps from funnel extractions to plate theory; the Craig section makes that jump visible.

drylab setup: Craig apparatus section (no preset needed — the controls are few). Two solutes with editable distribution ratios D, editable phase volumes, transfer count 0–200, per-tube results table, optional normal- approximation overlay.

Conceptual (in-class, groups):

- Set D = 4 and D = 0.25, equal phase volumes. “Before running: after one transfer, what fraction of each solute is in the mobile-phase tube? After two?” Groups compute the binomial split by hand for n = 2, then run at n = 2 and reconcile against the per-tube table.
- Increase transfers stepwise: 5 → 25 → 100. “Describe what happens to (a) the separation between the two bands and (b) the width of each band. Which grows faster?” Target invention: band centers separate linearly in n while widths grow as √n — the entire justification for columns, and the conceptual root of Rs ∝ √N in Exercise 3.
- Toggle the normal-approximation overlay at n = 100. “Where did the Gaussian come from — what assumption in the model produced it?” (Binomial → Gaussian; each transfer is an equilibration = one plate.)

Quantitative (out-of-class):

- For D = 1 and equal volumes, derive the tube of maximum concentration after n transfers (r_max = n·D/(D+1)) and the fraction in that tube; verify against the per-tube table at n = 50. Then predict r_max for D = 3 with V_upper = 2·V_lower (forces correct use of D vs. the volume-weighted partition ratio — a distinction the extraction learning objectives call out).
- Compute the minimum n for which bands with D = 1 and D = 4 reach Rs ≈ 1.5 using the √n width growth (centers separate as n·Δp, widths as 2√(np q) each), then find it empirically with the slider. Reconciling the estimate with the observed n *is* the assessment. (Revision 3 note: the pair was chosen so the answer lands inside the 200-transfer slider; closer D pairs push n into the hundreds — itself worth mentioning as the cost of small selectivity.)

Exercise 2 — Building the van Deemter curve before naming it

Wenzel linkage: In-Class Problem Set #2 (longitudinal diffusion, mass transport broadening) and the "draw h vs. v and rationalize it" objectives. Wenzel develops each broadening term as a separate thought experiment; here students *discover the shape empirically* first, then attach his mechanisms to its limbs.

drylab setup: HPLC section, Default preset, **Sweeps tab**. Sweep parameter = flow_rate, output = hetp. Default column (Zorbax SB-C18, 100 × 4.6 mm, 3 µm).

Conceptual (in-class, run *after* Wenzel's problem set #2, as its consolidation):

- Before plotting: "You've argued that longitudinal-diffusion broadening falls with flow rate and mass-transport broadening rises with it. Sketch h vs. v." Then run the sweep. Groups annotate the curve with *which mechanism owns which limb* and mark the optimum.
- Temperature probe: re-run the sweep at 10 °C and at 70 °C (temperature slider, then re-open the Sweeps tab). "The minimum moved. Which broadening term is temperature acting through, and how?" (D_m rises with T → B-limb rises, C-limb falls, optimum shifts right — connects Wenzel's diffusion problems to an observable.)
- The tradeoff question Wenzel asks verbatim ("why are most analyses not run at the optimum flow rate?"): switch the sweep output to t_r over the same range and have groups defend an operating point using both plots. *Instructor note:* at the default conditions the minimum sits at ~1.6 mL/min with h_min ≈ 6.0 µm (h ≈ 15.4 µm at 0.2 mL/min, 8.1 µm at 5.0 mL/min; measured at 50 %B MeCN, 25 °C, default column) — so the default 2.0 mL/min is *already near-optimal*, and the honest answer here is "we nearly do, at this particle size"; the classic run-fast-above- optimum tradeoff appears when they lower the temperature or raise the particle size, both of which they can test.

Quantitative (out-of-class):

- In the Sweeps tab, press **Sweep data (CSV)** — the file carries the swept parameter, the output, and the operating-point stamp. Fit $h = A + B/v + C \cdot v$ (any tool) and report A, B, C with units; compute $v_{opt} = \sqrt{B/C}$ and $h_{min} = A + 2\sqrt{BC}$ from the fit and compare with the plotted minimum. This converts "write an appropriate term for the van Deemter equation" (a stated learning objective) into an operational skill. (The sweep's x is flow rate, not linear velocity; either fit in flow-rate units or convert via the void time in the Results tab's System table — deciding which, and saying so, is part of the exercise.)
- Particle-size epilogue: set the particle slider from 3 µm to 6 µm, re-sweep, re-fit. "Report how A, B, C each moved, then check the backpressure in the System table. State the tradeoff in one sentence." (Ties to Homework 1, question 3c, and sets up why UHPLC exists — h_min scales with d_p but ΔP with $1/d_p^2$.)

Exercise 2A — One mechanism at a time: the van Deemter term builder

Wenzel linkage: the heart of In-Class Problem Set #2, which develops eddy diffusion, longitudinal diffusion, and the two mass-transfer terms as *separate* thought experiments before

ever summing them. Exercise 2 meets the curve as an emergent property of a full instrument model; this exercise builds it limb by limb first. Run 2A immediately before 2.

drylab setup: Band shape → Mechanisms (van Deemter) tab. One slider per mechanism: particle size d_p and packing factor λ (the A term), diffusion coefficient D_m (the B term; γ fixed at 0.6 and labeled), and the two C coefficients C_s and C_m entered directly in ms. Each term is its own curve under the bold total; the diamond marks the closed-form optimum u_opt = √(B/C), H_min = A + 2√(BC).

Conceptual (in-class):

- Zero everything you can: set C_s = C_m = 0 and note what remains and how it varies with u (A flat, B/u falling — and no optimum exists; the operating-point stamp says so). “Why does an open-tubular GC column have no A term?” (No packing — nothing to take multiple paths around; Wenzel’s LO on open-tubular columns, now demonstrable by construction.)
- Restore C_s = 2 ms, C_m = 1 ms. Predict, then verify with the sliders: what does doubling d_p do to which limb? Halving λ (a better-packed column)? Doubling D_m (a hotter, thinner mobile phase)? The last one moves *two* limbs in opposite directions — B up, effective C down is NOT what this builder does (C is entered directly), and noticing that the builder’s C sliders don’t respond to D_m is the setup for the question: “which physical variables does each C term actually contain?” Groups write the C_s and C_m expressions from lecture (d_f^2/D_s, d_p^2/D_m) and state which slider in the *HPLC* section would move them.
- Sum question: “Two columns have the same H_min but different shapes. Which would you rather run fast, and what number tells you?” (The C sum — flat curves forgive speed; this is the same reading as the GC Golay tab’s carrier comparison, and saying so links the sections.)

Quantitative (out-of-class):

- Export **Term curves (CSV)**. Verify numerically that the total’s minimum sits at √(B/C) and equals A + 2√(BC), using A, B, C recovered from the exported columns (A is the flat column; B from B/u · u at any point; C from the slope of the rising limbs). This is the same identity Exercise 2’s fit will extract from a *simulated instrument* — done here on curves the student built, so nothing is hidden.
- Sketch (no simulator): the same decomposition for an open-tubular GC column. Which limbs survive, and what replaces d_p in the C_m term? (Golay: A = 0, C_m carries d_c^2/D_m — bridges to Exercise 5’s coda.)

Exercise 3 — Triage with the fundamental resolution equation

Wenzel linkage: In-Class Problem Set #3 verbatim (“examine a chromatogram with unacceptable resolution and determine which term of the three is the best one to alter”), plus the k′-tradeoff objectives.

drylab setup: HPLC section, **preset “Ex 3 — Resolution triage”** (`?hplc=ex3`): acetophenone + p-chlorophenol at 50 %B acetonitrile on the default column. Opens at **Rs = 0.65, α = 1.031** — visibly fused, honestly salvageable. The **Rs (vs prev)** and **alpha (vs prev)** columns in the Results tab adjudicate every attempt live.

Conceptual (in-class):

- Present the failing chromatogram. Groups get one “budget line” each: **N-team** may only change length/particle size/flow; **k-team** only %B/temperature; **α-team** only the solvent (MeCN ↔ MeOH). Each team predicts *in writing* what its lever will do to Rs, tR, and backpressure, then tests it. Whole-class reconciliation ranks the levers. *Instructor key (all measured on the preset, other conditions default):* the N-team maxing **both** sliders (300 mm + 1.5 µm) reaches only **Rs = 1.56 at 2082 bar** — a number no pump survives, and the √N lesson embodied; the k-team at 44 %B gets **Rs = 2.58** in one move; the α-team with methanol at 65 %B gets **Rs = 7.79 (α = 1.57)**. Chemistry wins.
- *Instructor note, worth stating rather than letting groups discover an inconsistency:* on this pair, %B moves α as well as k (α = 1.031 at 50 %B → 1.115 at 44 %B), because the two solutes’ S coefficients differ. That is real reversed-phase behavior: the “three independent levers” of the fundamental equation are a *decomposition*, not a promise that one knob touches one term.
- Diminishing-returns probe for the k-team: Sweeps tab, k_prime vs. solvent_percent. “Where does further retention stop buying resolution?” Target: the k/(1+k) factor saturates — retention past k ≈ 5–10 costs time, not separation (Wenzel’s “limits to the effect of increasing the retention factor”).

Quantitative (out-of-class):

- From the Results table at the preset condition, record N (System table), α and the tabulated Rs. Compute Rs from the fundamental equation and compare with the table’s value — they will not match exactly, because the table’s Rs is built from the model’s full peak widths, extra-column broadening included; explaining the discrepancy is part (b). The **Export tab’s compound-results CSV** carries `rs_vs_prev` and `alpha_vs_prev` with the operating-point stamp, so the homework travels with its evidence.
- Predict Rs and tR if the column is lengthened from 100 mm to 300 mm (Rs ∝ √N, t ∝ L), then verify with the length slider. One-sentence verdict: when is tripling the column *ever* the right first move?

Exercise 4 — The general elution problem, twice: gradient LC and programmed GC

Wenzel linkage: In-Class Problem Set #3 problem 2 — the chromatogram with crowded early peaks and flat late peaks; “propose a way in gas chromatography / in liquid chromatography to eliminate both problems.” drylab shows both proposals working, side by side, which no static problem set can.

drylab setup: (a) HPLC, **preset “Ex 4 — General elution”** (`?hplc=ex4`): all 22 compounds at 30 %B. (b) GC, **preset “Ex 4 — General elution”** (`?gc=ex4`): the alkane ladder Heptane→Pentadecane on SLB-5ms at 120 °C, using the Programmed and Compare tabs.

Conceptual (in-class):

- LC leg: run the preset as-is at 30 %B, then push the slider to 70 %B. “Neither works. State, using the term from lecture, why not.” *Instructor key (measured, default column):* 30 %B

spans k′ 1.3–352 with the last peak near **188 min**; 70 %B compresses everything to k′ 0.22–5 inside 3 min — the two failure modes of the general elution problem in one slider throw. Then hand over the gradient editor (Gradient tab): groups design a program that gets every k′ into the useful window. The **Band tracks** view (k′ vs. time) is the payoff — students watch each band's k′ collapse as the gradient front reaches it, the mechanism Wenzel's text describes in words.

- GC leg: same logic with the oven. Isothermal 120 °C (17.4 min for nine alkanes, early ones crowded, C15 flat and late) vs. a group-designed ramp on the **Compare** tab, which holds column, carrier and pneumatics fixed so the two traces provably differ only in the elution program. "Why does the programmed run *sharpen* the late peaks rather than just moving them?" (Late solutes sit parked near the inlet until the oven catches up — the band-compression argument.)
- Synthesis question, answered without the sim: "State in one sentence what a %B-gradient and temperature-programming have in common." (Both ramp the effective k′ of everything downward during the run.)

Quantitative (out-of-class):

- Dwell-volume forensics (LC): change the mixing/non-mixing volumes in the Gradient tab and measure the shift in early-peak retention against the Solvent-profile view. Compute the expected delay t_dwell = V_dwell/F and compare. (Instrument realism that transfers directly to lab hardware.)
- Peak-capacity estimate (GC): from the **Peak results (CSV)** exports of the isothermal and programmed runs (both arms have export buttons under their tables), compute $n_c \approx 1 + (\sqrt{N}/4)\cdot\ln(t_{last}/t_{first})$ for the isothermal run vs. the gradient-mode estimate for the program, and report the ratio. The number makes "programming fixes the general elution problem" quantitative.

Exercise 5 — What retains a molecule in GC: boiling point, polarity, and ln k vs. 1/T

Wenzel linkage: In-class GC problems (pentane vs. pentanol boiling points, retention-order prediction), Homework 2 problem 1 (esters on polar vs. non-polar phases: does *order* change? do *times* change?), and the "predict the retention order in GC" objective.

drylab setup: GC isothermal, three presets in sequence, plus the **Golay tab** for the carrier-gas coda. The solute picker keeps each round to the compounds under discussion.

Conceptual (in-class):

- Round 1 (volatility axis), **preset "Ex 5 — Alkane ladder"** (`?gc=ex5a`): Octane, Decane, Dodecane on SLB-5ms at 120 °C. Predict the order — easy — then: "Retention doesn't just increase, it increases *multiplicatively* per CH_2. Propose why." (Free energy is additive per methylene → k is multiplicative; the molecular version of the ln k linearity they'll measure below.)
- Round 2 (polarity axis), **preset "Ex 5 — Polarity trio"** (`?gc=ex5b`): Decane (bp 174 °C), 1-Octanol (bp 195 °C), 2-Octanone on SLB-5ms; groups predict, run, then switch the phase

dropdown to Wax and predict again before running. *Instructor key (measured at 120 °C, default column):* on SLB-5ms the order is **2-Octanone < Decane < 1-Octanol**; on Wax it becomes **Decane < 2-Octanone < 1-Octanol** — the alkane vaults from the middle to first out as the polar phase releases it, while the H-bonding alcohol stays anchored last on both. “Which property set the order on each phase? Whose *absolute* retention changed most?”

- Carrier-gas coda (bridges back to Exercise 2): open the **Golay tab** — plate height vs. average carrier velocity for one solute, all three gases on one axis, diamonds marking each gas’s optimum, swept over the inlet-pressure control’s own range. “Wenzel’s problem set says longitudinal diffusion matters more in gases. Read the plot: which carrier suffers most at low velocity, which has the flattest fast side, and which would you buy?” (N_2’s low sharp optimum against H_2’s flat fast curve is the textbook argument, now read directly off the figure; the **Golay sweep data (CSV)** button exports the curves.)

Quantitative (out-of-class):

- Pick one solute; record k from the results table at four oven temperatures spanning ~40 °C (the **Peak results (CSV)** export stamps each file with its temperature). Plot ln k vs. 1/T, extract the slope, and interpret it as $-\Delta H°/R$ for transfer into the stationary phase. Compare the ΔH ranking of Decane vs. 1-Octanol on Wax with Round 2’s conclusion. (The K-centric model the sim runs on makes this linearity real, not decorative.)
- Co-elution hunt, **preset “Ex 5 — Co-elution hunt”** (`?gc=ex5c`): Tridecane and 2-Octanone on Wax, opening at 120 °C where the alkane elutes first. From two 4-point ln k vs. 1/T fits, predict the temperature at which the pair co-elutes, then verify by walking the oven slider down. *Instructor key (measured):* the ln k lines cross near **102 °C** at k ≈ 1.7 — comfortably observable — and the order is inverted on the far side. A co-elution temperature as an *intersection of thermodynamic lines* is a high-value concept the static problem sets cannot reach.

Exercise 6 — Why the CE detector sees everything: EOF as a controllable variable

Wenzel linkage: CE In-Class Problems 1–12 nearly verbatim — silanol chemistry, EOF vs. pH, the “anions, cations and neutrals are all detected — explain” question, and the five-substance electropherogram prediction. Also the CE out-of-class amino-acid question.

drylab setup: CZE section (no preset — the defaults are the scenario). The EOF-mobility slider (0–80 $\times 10^{-9}$ m²/Vs) is the pedagogical control: it is exactly the “pH knob” of Wenzel’s problems 2 and 7, since bare-silica EOF falls toward zero at low pH. The default composition already carries K^+, Na^+, Li^+ over acetate/Tris.

Conceptual (in-class):

- EOF = 0 (a “pH 2.5 capillary”): run. Only cations reach the detector. Groups explain, then predict what EOF = 60 $\times 10^{-9}$ (a “pH ~9 capillary”) changes — committing the predicted arrival order of a cation, a neutral, and an anion to paper — then run and reconcile. This *is* Wenzel problem 11, with the mechanism exposed as a slider instead of asserted.

- K^+ / Na^+ / Li^+ order: “All +1. Crystal-radius order says Li^+ is smallest — why is it slowest?” (Hydrated radius vs. crystal radius — mirrors the ion-exchange retention-order reasoning in Problem Set #4, a deliberate echo worth naming for the class.)
- Broadening callback: “Of the band-broadening mechanisms from the van Deemter work — eddy, longitudinal, stationary-phase and mobile-phase mass transport — which survive in CE?” Then raise the voltage until the peaks visibly degrade and ask what new mechanism appeared (Joule heating; the sim reports it).

Quantitative (out-of-class):

- From μ_app = μ_ep + μ_EOF, predict the migration times of K^+, Na^+ and Li^+ at the defaults (500 V over 30 mm; detector at 15 mm) — noting that the injected plug is centered at 6 mm, so the migration distance is 9 mm, not 15 — and compare with the simulated peak times. The simulated times run ~10–15 % longer than infinite-dilution arithmetic because the model computes effective mobilities in the full electrolyte; explaining that systematic (order and ratios right, absolute times slower) is part of the answer, not an error to hide. Then the acetate problem: in the default composition acetate is a **BGE** ion — it fills the capillary and shows no peak, so first switch its role to Sample in the composition editor (realizing *why* nothing eluted is part of the exercise). With the plug at 6 mm, compute the minimum EOF for acetate (μ_ep = −42.4) to reach the detector within the default 20 s run, and the (smaller) minimum for a 60 s run; verify both by bracketing with the slider.
- Resolution vs. voltage: N in CE grows linearly with V (until heating). Run three voltages, compute plate counts for one peak from the reported migration time and FWHM ($N = 5.54(t/w_{½})^2$ — deliberately left as hand arithmetic), identify where linear scaling breaks, and correlate with the Joule-heating report.

Exercise 7 — When the column is full: overload, tailing, and fronting

Wenzel linkage: the peak-shape and column-capacity discussion — why real peaks tail, what “overload” means thermodynamically, and why an asymmetric peak still carries information. This is the exercise the unit could not previously run: no static problem set can show an isotherm *bending* a peak.

drylab setup: Band shape → Peak shape (overload) tab, at its defaults (N = 100, k = 2, curvature 0, load 1). The dashed curve is the exact linear-isotherm profile at the same N and k — the honest baseline — and the readout table reports t_max, As at 10 % height (for both the simulated and the ideal profile), apparent vs. built-in plate count, and mass recovery. Time is in transfers (one transfer = one plate volume); the reduced units are the point: every effect shown is scale-free.

Conceptual (in-class):

- Baseline literacy first: at the defaults the simulated and ideal curves coincide, and the *ideal* profile’s own As reads ≈ 1.18, not 1.00. “Nothing is overloaded — where is the asymmetry coming from?” (A 100-plate column is a 100-event random process; the plate model’s profile is skewed at finite N and only becomes Gaussian as N grows. Raise N toward 1000

and watch As_ideal fall toward 1. Asymmetry only means something against this baseline — which is why the table prints both.)

- Commit predictions, then push **curvature positive** (Langmuir, the isotherm bending toward saturation): which side of the peak sharpens, and does t_max move early or late? Then negative. Reconcile with the isotherm argument: velocity of a concentration goes as 1/(1 + F·dq/dC); Langmuir's dq/dC *falls* with C, so the top of the band outruns the base — sharp front, diffuse rear (tailing), maximum arriving earlier. Anti-Langmuir is the mirror (fronting, maximum later).
- The two tailings: set curvature back to 0 and raise the **slow-site τ** slider instead. The peak tails again — with t_max moving *later*, not earlier, and at *any* load. "Your real column shows a tailing peak. Name an experiment that tells overload tailing from slow-site tailing." (Cut the load: overload tailing dies, kinetic tailing doesn't — and the t_max direction differs. This is a genuinely diagnostic pair of observables.)
- Load ladder: curvature 0.5, load 1 → 4 → 10. Watch As climb, t_max walk earlier, and apparent N fall below the built-in 100. "The column didn't change — why did it 'lose' plates?" (It didn't; the moment-based plate count assumes a symmetric peak, and overload broke the assumption. What a number *means* depends on the model behind it.)

Quantitative (out-of-class):

- Export the **Profile (CSV)** at curvature 0, +2, and −0.6 (load 1, N = 100, k = 2). For each, compute As at 10 % height from the exported points and the first-moment retention; tabulate against the on-screen readouts. Then state, from the +2 profile alone, whether the *front* or the *rear* of the band carries the isotherm information, and why the tail of a Langmuir-overloaded peak points at the linear-isotherm retention time.
- Design question (no simulator): a QC method reports "plate count" from a moment fit and flags the column as degraded whenever N_app drops 20 %. The instrument is also used for a high-load prep separation. Explain, with the numbers from the load ladder, how this flag can fire on a healthy column — and what second readout (As, and t_max direction) distinguishes the cases.

Coverage map and honest gaps

Wenzel unit topic	drylab exercise	Status
Extraction / partition	Ex. 1 (Craig)	Conceptual bridge; wet-lab schemes stay on paper
Longitudinal diffusion, mass transport, van Deemter/Golay	Ex. 2A, Ex. 2, Ex. 5 coda (Golay tab)	Full
Eddy diffusion	Ex. 2A (Mechanisms tab)	Full since v0.24.0: the A-term is a live control (λ, d_p); its absence in open-tubular GC is demonstrated by construction
Fundamental resolution equation	Ex. 3	Full, with live Rs/α columns
General elution problem / gradients / T-programming	Ex. 4	Full

Wenzel unit topic	drylab exercise	Status
GC retention order & thermodynamics	Ex. 5	Full, incl. a verified co-elution crossing
Steric exclusion, ion exchange	—	Not simulatable in drylab; Wenzel's paper problems stand alone (Ex. 6 echoes the IEC hydrated-radius argument)
CE	Ex. 6	Full (CZE; no ITP/stacking, consistent with the ruled CE scope)
Peak asymmetry: overload (isotherm curvature) and kinetic tailing	Ex. 7 (Peak shape tab)	Full since v0.24.0; the two tailing mechanisms are separate controls and taught apart

Sequencing suggestion: Ex. 1 immediately after the extraction unit; Ex. 2A then Ex. 2 as the consolidation of In-Class Set #2; Ex. 7 directly after Ex. 2 while the plate picture is fresh (or with the GC block); Ex. 3–4 with Set #3; Ex. 5 with the GC lecture block; Ex. 6 with the CE block. Each in-class leg is sized for one 50-minute facilitated session in Wenzel's 3–4-student groups; each quantitative leg is one homework problem with its CSV evidence attached — every export drylab produces is stamped with the operating point that generated it, so "attach your CSV" is also "state your conditions."

Assessment rubric hook (all exercises): grade the *prediction–reconciliation delta*, not the final answer — full credit requires the written pre-simulation prediction, the observed result, and an explicit statement of what changed in the student's model when they disagreed. That is the POGIL contract, and it is what distinguishes these from "play with the simulator" exercises.